

Number Sense

Marking Scheme & Blueprint

MTH1W · Module 1 Test — Teacher Copy

Blueprint

Curriculum: MTH1W (Grade 9 de-streamed math). Total: 70 marks · 60 minutes · 30 questions. Categories: K/U, Application (App), Thinking (Think), Communication (Comm).

Q	Sub-topic	Level	Marks	Category	Section
A1	Integers	L1	1	K/U	A
A2	Integers	L1	1	K/U	A
A3	Order of Operations	L1	1	K/U	A
A4	Exponent Rules	L1	1	K/U	A
A5	Exponent Rules	L1	1	K/U	A
A6	Fractions	L1	1	K/U	A
A7	Fractions	L1	1	K/U	A
A8	Decimals/Percent	L1	1	K/U	A
A9	Ratios & Rates	L1	1	K/U	A
A10	Order of Operations	L1	2	K/U	A
A11	Fractions	L1	2	K/U	A
A12	Applying Percents	L1	2	K/U	A
A13	Square Roots & Irrationals	L1	2	K/U	A
A14	Decimals/Percent	L1	2	K/U	A
A15	Ratios & Rates	L1	2	K/U	A
B16	Fractions	L2	2	App	B
B17	Exponent Rules	L2	2	App	B
B18	Applying Percents	L2	2	App	B
B19	Rational Numbers	L2	2	App	B
B20	Ratios & Rates	L2	2	App	B
B21	Square Roots & Irrationals	L2	2	App	B
B22	Fractions	L2	3	App	B
B23	Applying Percents	L2	3	App	B
B24	Ratios & Rates	L2	3	App	B
C25	Exponent Rules	L3	4	Think	C
C26	Applying Percents	L3	4	Think	C
C27	Applying Percents (mixture)	L3	5	Think	C
C28	Applying Percents (chain)	L3	5	Think	C
C29	Fractions (work backwards)	L3	6	Think	C
D30	Square Roots & Irrationals	L3	4	Comm	D

By category (marks): K/U 21 · Application 21 · Thinking 24 · Communication 4

By level (marks): L1 21 · L2 21 · L3 28 → L1 30% · L2 30% · L3 40%

Worked Solutions & Mark Allocation

- 1. (A1 · K/U · 1 marks)**
= -16. (A1)
- 2. (A2 · K/U · 1 marks)**
= -20 + 8 = -12. (A1)
- 3. (A3 · K/U · 1 marks)**
= 6 + 20 = 26. (A1)
- 4. (A4 · K/U · 1 marks)**
 $3^{2+3} = 3^5$. (A1)
- 5. (A5 · K/U · 1 marks)**
 $m^{8-5} = m^3$. (A1)
- 6. (A6 · K/U · 1 marks)**
 $2/8 + 3/8 = 5/8$. (A1)
- 7. (A7 · K/U · 1 marks)**
= $18/30 = 3/5$. (A1)
- 8. (A8 · K/U · 1 marks)**
 $0.08 \times 100 = 8\%$. (A1)
- 9. (A9 · K/U · 1 marks)**
 $\div 12: 2 : 3$. (A1)
- 10. (A10 · K/U · 2 marks)**
 $(5)^2=25$; $25 \div 5=5$; $40-5 = 35$. (M1, A1)
- 11. (A11 · K/U · 2 marks)**
 $5/6 - 3/6 = 1/3$. (M1 common denom, A1)
- 12. (A12 · K/U · 2 marks)**
 $0.12 \times 150 = 18$. (A2)
- 13. (A13 · K/U · 2 marks)**
 $\sqrt{49} = 7$ rational; $\sqrt{50}$ irrational. (A1, A1)
- 14. (A14 · K/U · 2 marks)**
 $5 \div 8 = 0.625 = 62.5\%$. (A1, A1)
- 15. (A15 · K/U · 2 marks)**
 $180 \div 3 = 60$ km/h. (M1, A1)
- 16. (B16 · Application · 2 marks)**
 $3/4 \times 3/2 = 9/8$. (M1 reciprocal, A1)

- 17. (B17 • Application • 2 marks)**
 $(2^2)^3 = 2^6$; $2^6 \div 2^4 = 2^2 = 4$. (M1, A1)
- 18. (B18 • Application • 2 marks)**
 $80 \times 1.13 = \$90.40$. (M1, A1)
- 19. (B19 • Application • 2 marks)**
 $-2/3 \approx -0.667$, $-1/2 = -0.5$.
 Order: $-2/3$, -0.6 , -0.55 , $-1/2$. (M1, A1)
- 20. (B20 • Application • 2 marks)**
 Unit rate $12.50 \div 5 = \$2.50/\text{kg}$; $8 \times 2.50 = \$20.00$. (M1, A1)
- 21. (B21 • Application • 2 marks)**
 $7^2 = 49 < 60 < 64 = 8^2$, so between 7 and 8. (M1, A1)
- 22. (B22 • Application • 3 marks)**
 Multiply first: $1/4 \times 8/5 = 2/5$.
 $2/3 + 2/5 = 10/15 + 6/15 = 16/15$. (M1, M1, A1)
- 23. (B23 • Application • 3 marks)**
 Increase = 50.
 $50 \div 250 \times 100 = 20\%$. (M1, M1, A1)
- 24. (B24 • Application • 3 marks)**
 Parts = 12; one part = $840 \div 12 = \$70$.
 $3 \times 70 = \$210$, $4 \times 70 = \$280$, $5 \times 70 = \$350$. (M1, A1, A1)
- 25. (C25 • Thinking • 4 marks)**
 $(a^2b^3)^2 = a^4b^6$ (power of a product / power of a power). (M1)
 $a^4b^6 \div (a^1b^2) = a^{4-1}b^{6-2}$ (quotient rule). (M1)
 $= a^3b^4$. (A1 result, C1 laws named)
- 26. (C26 • Thinking • 4 marks)**
 Combined factor = $1.25 \times 0.80 = 1.00$. (M1)
 $1.00 \times p = 120 \rightarrow p = \120 . (M1, A1)
 Net effect: the two changes cancel — final price equals the original. (A1)
- 27. (C27 • Thinking • 5 marks)**
 Acid now = $0.20 \times 750 = 150$ mL. Let x = pure acid added. (M1)
 $(150 + x)/(750 + x) = 0.40 \rightarrow 150 + x = 300 + 0.4x$. (M1)
 $0.6x = 150 \rightarrow x = 250$ mL of pure acid. (M1, A1, A1)
- 28. (C28 • Thinking • 5 marks)**
 Combined factor = $1.20 \times 0.85 = 1.02$. (M1, M1)
 $1.02 \times p = 102 \rightarrow p = 102 \div 1.02 = 100$. (M1, A1, A1)

29. (C29 • Thinking • 6 marks)

Let C = capacity. $(\frac{2}{5})C + 24 = (\frac{4}{7})C$. (M1)

$24 = (\frac{4}{7} - \frac{2}{5})C = (\frac{20}{35} - \frac{14}{35})C = (\frac{6}{35})C$. (M1, M1)

$C = 24 \times 35 \div 6 = 140$ L. (M1, A1)

Check: $\frac{2}{5} \times 140 = 56$; $56 + 24 = 80 = \frac{4}{7} \times 140$. ✓ (A1)

30. (D30 • Communication • 4 marks)

$\sqrt{4} + \sqrt{9} = 2 + 3 = 5$, but $\sqrt{13} \approx 3.61$. (A1 calculation)

Since $5 \neq \sqrt{13}$, Ravi is not correct. (A1 conclusion)

General rule: the square root does not distribute over addition; $\sqrt{a} + \sqrt{b} \neq \sqrt{a+b}$ in general. (C1, C1 communication)